

Introduction to Contract Theory and Auctions

Order Statistics & Stochastic Dominance

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- Let $X_1, X_2, \dots, X_n$ be n independent draws from a distribution F with associated density f , i.e., $(X_i)_{i=1, \dots, n}$ is a **random sample: i.i.d. with $X_i \sim F, i = 1, \dots, n$** .
- Let $Y_1^{(n)}, \dots, Y_2^{(n)}, \dots, Y_n^{(n)}$ be a rearrangement:

$$Y_1^{(n)} \geq Y_2^{(n)} \geq \dots \geq Y_n^{(n)}.$$

- The random variables $Y_k^{(n)}, k = 1, \dots, n$ are called the **Order Statistics**.
- Let $G_k^{(n)}$ denote the distribution of $Y_k^{(n)}$, with corresponding density $g_k^{(n)}$.
- In our study of auctions, we will frequently encounter the highest and second-highest order statistics: $Y_1^{(n)}$ and $Y_2^{(n)}$.
- Note: when the sample size n is fixed and there is no ambiguity, I will suppress the notation (n) and just write Y_k instead of $Y_k^{(n)}$, and similarly for the distribution G_k instead of $G_k^{(n)}$ and the density g_k instead of $g_k^{(n)}$.

Highest order statistic Y_1 .

- The event $Y_1 \leq y$ is the same as the event: for all k , $X_k \leq y$. Since each $X_k \sim F$ i.i.d., we have that:

$$G_1(y) = F(y)^n, \quad g_1(y) = nF(y)^{n-1}f(y)$$

Second-highest order statistic Y_2 .

- The event that $Y_2 \leq y$ is the union of the following disjoint events:

$$\{X_k \leq y, \forall k = 1, \dots, n\}$$

OR

$$\{n-1 \text{ of the } X_k \text{ are less than or equal to } y \text{ and one is greater than } y.\}$$

- There are $\binom{n}{n-1} = n$ ways that the second event can happen, thus:

$$\begin{aligned} G_2(y) &= F(y)^n + nF(y)^{n-1}(1 - F(y)) \\ &= nF(y)^{n-1} - (n-1)F(y)^n \end{aligned}$$

- The density function is:

$$g_2(y) = n(n-1)(1 - F(y))F(y)^{n-2}f(y)$$

Relationships between the highest and second-highest order statistics.

Note that

$$\begin{aligned} G_2^{(n)}(y) &= nF(y)^{n-1} - (n-1)F(y)^n \\ &= nG_1^{(n-1)}(y) - (n-1)G_1^{(n)}(y) \end{aligned}$$

and thus, the density satisfies:

$$g_2^{(n)}(y) = ng_1^{(n-1)}(y) - (n-1)g_1^{(n)}(y)$$

The above immediately implies that

$$\mathbb{E}[Y_2^{(n)}] = n\mathbb{E}[Y_1^{(n-1)}] - (n-1)\mathbb{E}[Y_1^{(n)}]$$

We also have that:

$$\begin{aligned} g_2^{(n)}(y) &= n(n-1)(1-F(y))F(y)^{n-2}f(y) \\ &= n(1-F(y))g_1^{(n-1)}(y) \end{aligned}$$

Joint and Conditional Distributions of Order Statistics

- Recall that although $X_1, \dots, X_n$ are independent, the order statistics are not.
- Let $\mathbf{Y} = (Y_1^{(n)}, \dots, Y_n^{(n)})$ be the vector of order statistics.
- Then the **joint density** of $\mathbf{Y}$ is

$$g_{\mathbf{Y}}(y_1, \dots, y_n) = \begin{cases} n! f(y_1) f(y_2) \cdots f(y_n) & \text{for } y_1 \geq y_2 \geq \cdots \geq y_n \\ 0 & \text{otherwise.} \end{cases}$$

- From here we deduce¹ that the joint density of the first- and second-order statistics is

$$g_{1,2}^{(n)}(y_1, y_2) = \begin{cases} n(n-1) f(y_1) f(y_2) F(y_2)^{n-2} & y_1 \geq y_2 \\ 0 & \text{otherwise} \end{cases}$$

¹This is done by integrating out all the other variables using the appropriate ranges of integration. You will need to use integration by parts many times.

Conditional distributions

- The density of $Y_2^{(n)}$ conditional on $Y_1^{(n)} = y$ is

$$\begin{aligned} g_2^{(n)}(z|Y_1^{(n)} = y) &= \frac{g_{1,2}^{(n)}(y, z)}{g_1^{(n)}(y)} \\ &= \frac{n(n-1)f(y)f(z)F(z)^{n-2}}{nF(y)^{n-1}f(y)} \\ &= \frac{(n-1)f(z)F(z)^{n-2}}{F(y)^{n-1}} \end{aligned}$$

if $y \geq z$ and 0 otherwise.

- The density of $Y_1^{(n-1)}$ conditional on the event $Y_1^{(n-1)} < y$ is

$$g_1^{(n-1)}(z|Y_1^{(n-1)} < y) = \frac{g_1^{(n-1)}(z)}{G_1^{(n-1)}(y)} = \frac{(n-1)f(z)F(z)^{n-2}}{F(y)^{n-1}}$$

- Thus $g_2^{(n)}(z|Y_1^{(n)} = y) = g_1^{(n-1)}(z|Y_1^{(n-1)} < y)$ for all $z < y$

Comparison of Payoff distributions in terms of return and risk.

- In many applications in microeconomics, including auction theory, it will be important to be able to compare *payoff distributions*.
- In particular, we might want to be able to formalize the following question: When does a decision maker prefer a payoff distribution F over the payoff distribution G ?
- Let $Z \subseteq \mathbb{R}$ be a set of monetary prizes. In particular, assume that Z is either of the form $[w, b]$, with $w < b$ or $[w, \infty)$.
- Let $\Delta(Z)$ be the set of CDFs on Z . For $F \in \Delta(Z)$, the interpretation for $F(z)$ is the probability that the random reward, whose distribution is given by F , is less than the number z .
- We will try to formalize the answer the que question I posed by formalizing some orders (partial orders to be specific) on the set $\Delta(Z)$.

Notation for Integrals

In what follows, you may notice that when taking the expectation of a function $h(x)$ with respect to a cumulative distribution function (CDF) F , I write:

$$E[h] = \int h(x) dF(x)$$

The expression above is called a **Riemann-Stieltjes integral** (or a **Lebesgue-Stieltjes integral**, for the hardcore among you).

This type of integral unifies the treatment of expectations for discrete, continuous, and even more exotic random variables. So, don't be intimidated—it's simply a more general and powerful definition of the integral that works in all cases.

For the examples we'll cover in class, when F is the CDF of a continuous random variable that admits a probability density function (pdf) f , the integral simplifies to the familiar form:

$$E[h] = \int h(x) f(x) dx,$$

which is the standard expectation formula using the **Riemann integral** that you all know and allegedly love.

First Order Stochastic Dominance

- We want to formalize the expression: “The distribution F yields unambiguously higher returns than the distribution G ”.
- There are two possibilities:
 - We could test whether every expected utility maximizer who values more over less money, prefers F to G .
 - Alternatively, we could verify if for every amount of money x , the probability of getting at least x is higher under F than G .
- Turns out that both ways of looking at this issue are the same.

Definition A.1

Let $F, G, \in \Delta(Z)$. The distribution F **first-order stochastically dominates** G , denoted $F \succsim_1 G$, if, for every non-decreasing function $u : Z \rightarrow \mathbb{R}$, we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Proposition A.2

Let $F, G \in \Delta(Z)$. Then $F \succsim_1 G$ if and only if $F(x) \leq G(x)$ for every x .

Equivalently, define for any CDF F , $\bar{F}(z) = 1 - F(z)$ for every $z \in Z$. Intuitively, $\bar{F}(z) = \Pr(\text{the monetary prize is greater than } z)$. Then. $F \succsim_1 G \Leftrightarrow$ if $\bar{F}(x) \geq \bar{G}(x)$ for all x .

Proof about equivalence of FOSD definitions

Proof.

Let $F, G \in \Delta(Z)$. To simplify the proof, we will assume that the distributions F and G have pdf's f and g respectively. Note that it suffices to establish the result for differentiable utility functions u . Recall that we assumed $Z = [w, b] \subset \mathbb{R}$, where $w < b$. We establish an equivalent form for $E[u(F)]$ first:

$$\begin{aligned}\int_w^b u(z) dF(z) &= \int_w^b u(z) f(z) dz = \int_w^b \left(\int_w^z u'(x) dx + u(w) \right) f(z) dz \\ &= \int_w^b \left(\int_w^z u'(x) dx \right) f(z) dz + u(w) = \int_w^b \int_x^b u'(x) f(z) dz dx + u(w) \\ &= \int_w^b u'(x) \left(\int_x^b f(z) dz \right) dx + u(w) = \int_w^b u'(x) \bar{F}(x) dx + u(w)\end{aligned}$$



Proof.

We prove the $\Leftarrow$ (if) part. Note that:

$$\int_w^b u(z) dF(z) \geq \int_w^b u(z) dG(z) \Leftrightarrow \int_w^b u'(x) \overline{F}(x) dx \geq \int_w^b u'(x) \overline{G}(x) dx.$$

Thus, if $\overline{F}(x) \geq \overline{G}(x)$ and since u is increasing (and thus $u' \geq 0$), then $\int_w^b u'(x) \overline{F}(x) dx \geq \int_w^b u'(x) \overline{G}(x) dx$ and the if part of the proposition is true.

We now prove the $\Rightarrow$ (only if) part. We use the contrapositive here. Suppose that there exists a prize $\hat{z}$ for which $\overline{F}(\hat{z}) < \overline{G}(\hat{z})$. Define the following utility function $\hat{u}(x) = 1$ for $x > \hat{z}$ and $\hat{u}(x) = 0$ for $x \leq \hat{z}$. This u is non-decreasing. But then $\int_w^b \hat{u}(x) dF(x) = \overline{F}(\hat{z}) < \overline{G}(\hat{z}) = \int_w^b \hat{u}(x) dG(x)$. Thus, by the contrapositive, the assertion is proved. □

Example A.3

Consider a compound rewards lottery that has as its first stage a realization of x according to G and in its second stage applies to the outcome x of the first stage an *upwards probabilistic shift*. That is, if outcome x is realized in the first stage, then the second stage pays a final amount of money $x + z$, where z is distributed according to $H_x(z)$ with $H_x(0) = 0$. Thus H_x generates a final return of at least x with probability one.

Denote the resulting distribution by F . Then for any nondecreasing $u : Z \rightarrow \mathbb{R}$ we have

$$\int u(t) dF(t) = \int \left(\int u(x + z) dH_x(z) \right) dG(x) \geq \int u(t) dG(t).$$

Thus, $F \succsim_1 G$.

- We want now to introduce a comparison based on relative riskiness or dispersion.

Definition A.4

For any two distributions F and G with the same mean, F **second-order stochastically dominates** G , denoted $F \succsim_2 G$, if for every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have

$$\int u(z) dF(z) \geq \int u(z) dG(z)$$

Definition A.5

For any $F, G \in \Delta(Z)$, G is a mean-preserving spread of F if and only if

$$Y = X + \epsilon$$

for some random variables $X \sim F$, $Y \sim G$ and ϵ such that $E[\epsilon|X = x] = 0$ for all x .

- Imagine that for every realization of x , we add a noise ϵ and give the decision maker $y = x + \epsilon$. In other words, we are spreading the probabilities without changing the mean.
- If the decision maker is risk averse, he would not like to have this scheme. He would rather have x .
- It is instructive to compare this to example A.3. In that example, we were giving the decision maker an extra amount of money for every realization x .
- While this could increase the variance of money, the decision maker knew that he was getting more.
- In the current scheme, we are increasing the variance without increasing the expectation.

Proposition A.6

Assume that $F, G \in \Delta(Z)$ have the same mean. Then, the following are equivalent.

- ① For every non-decreasing concave function $u : Z \rightarrow \mathbb{R}$, we have
$$\int u(z) dF(z) \geq \int u(z) dG(z)$$
- ② G is a mean-preserving spread of F .
- ③ For every $t \geq 0$, $\int_w^t G(x) dx \geq \int_w^t F(x) dx$.

Proof.

Homework! You can use a similar technique as in the proof of A.2, by using integration by parts twice. □

- Let $F, G \in \Delta(Z)$ be distributions with densities f and g respectively.
- Define the **odds-ratio**

$$I(x) = \frac{f(x)}{g(x)}.$$

- Suppose that you observe a draw x from a random variable that may have distribution F or G .
- You ask the question: *given the realization x , is it more likely that it came from F than from G ?*
- It turns out that $I(x)$ is the ratio to which you need to multiply the prior odds to get the posterior odds!

Definition A.7

The pair (f, g) has the MLRP (monotone likelihood ratio property) if $I(x)$ is non-decreasing.

- Intuitively, the higher the realized value of x , the more likely that it was drawn from the high distribution F .
- MLRP implies FOSD, but it is a stronger condition.
- You could have FOSD and still there might be some high signal values that likely come from G .

Example A.8

Suppose $f(0) = f(2) = 0.5$ and $g(1) = g(3) = 0.5$. Then $G \succsim_1 F$ but the MLRP property fails.

- The MLRP is often used as an assumption in models of moral hazard and screening.
- Let $F(x|a)$ be a family of distribution functions parameterized/indexed by the parameter a .
- The parameter a could be an action (e.g. effort), or a type (e.g. ability) of an agent, and x is the outcome (e.g. the amount produced/consumed).
- The MLRP tells us that if $x_2 > x_1$ and $a_2 > a_1$ then

$$\frac{f(x_2|a_2)}{f(x_2|a_1)} \geq \frac{f(x_1|a_2)}{f(x_1|a_1)}.$$

- In other words, if a principal observes a higher x , it will guess a higher likelihood that it came about due to a higher a .
- This is important in the design of incentives!